

Proposed New Division- KIPRE

Data Science and Analytics Division

a) Rationale

The Data Science and Analytics Division draws its mandate from Paragraph 5 of Legal Notice No. 273 of 2017, specifically focusing on functions (a), (c), and (g). It provides centralized, cross-cutting technical leadership in biostatistics, bioinformatics, and AI/ML modeling to drive data-driven research and evidence-based decision-making across all institutional programs.

b) Functions

The Division for Data Science and Analytics is charged with the following high-level macro functions:

- (i). Coordinate and provide cross-cutting biostatistical, bioinformatics, and computational science support across all research directorates and laboratory units.
- (ii). Design, implement, and maintain secure, reproducible, and scalable institutional workflows for high-dimensional omics, genomic, and clinical research data.
- (iii). Develop integrated data repositories, automated ingestion pipelines, and interactive visualization dashboards to inform disease surveillance and institutional interventions.
- (iv). Deploy advanced predictive modeling, artificial intelligence (AI), and machine learning (ML) frameworks to investigate disease transmission dynamics and risks.
- (v). Establish and enforce analytical data standards, model governance protocols, and quality assurance workflows in strict coordination with the ICT and data protection functions.
- (vi). Deliver specialized technical training, quantitative mentorship, and data science knowledge transfer across the institute's workforce.

Organizational Structure

The Data Science and Analytics Division will be headed by a Deputy Director, Data Science and Analytics and will comprise two technical sections, each headed by an Assistant Director:

1. *Biostatistics and Computational Analytics Section*
2. *Data Analytics and Decision Support Section*

1. Biostatistics and Computational Analytics Section

a) Rationale

The Biostatistics and Computational Analytics Section delivers centralized quantitative, epidemiological, and bioinformatics capabilities across all research programs. By deploying standardized computational pipelines and AI/ML modeling, the section translates high-dimensional multi-omics and biological data into actionable research insights, optimizing institutional resource utilization and eliminating structural duplication.

b) Functions

The Section is charged with the following specific functions:

- (i). Provide biostatistical, epidemiological, and computational support for all pre-clinical, clinical, and biological studies.
- (ii). Collect, process, and interpret genomic, pre and clinical, and epidemiological datasets alongside core research units.
- (iii). Deliver expert statistical advisory services to support multi-disease surveillance and evidence-based decisions.
- (iv). Execute GIS and spatial analysis to map the distribution, ecology, and transmission of zoonotic diseases.
- (v). Develop machine learning and predictive models to forecast disease patterns, risk factors, and intervention impacts.

- (vi). Design and maintain reproducible pipelines to process high-dimensional genomic, transcriptomic, and proteomic data.
- (vii). Conduct computational host-pathogen analytics to support biomedical, evolutionary, and ecological research portfolios.
- (viii). Build standardized computational frameworks and analytical tools to secure data integrity across the institute.
- (ix). Integrate diverse multi-dimensional biological datasets to discover molecular and disease-related patterns.
- (x). Enforce data quality, reproducibility, and interoperability standards across all quantitative research projects.
- (xi). Deliver technical training and capacity building in biostatistics, GIS, and data science for institutional staff.
- (xii). Define analytical standards, protocols, and operational best practices for quantitative workflows in all labs.
- (xiii). Partner with national and global institutions to deploy innovative computational methods that advance research.

2. Data Analytics and Decision Support Section

a) Rationale

The Data Analytics and Decision Support Section establishes enterprise-wide data engineering and operational intelligence systems. It architects automated data ingestion pipelines and secure analytical repositories, transforming diverse scientific and operational streams into interactive decision dashboards to drive real-time surveillance and institutional oversight.

b) Functions

- (i). Develop and coordinate analytical data standards, data access workflows, data quality procedures, and AI/ML model governance protocols in strict coordination with ICT and data protection functions.
- (ii). Build and maintain optimized analytical data repositories, metadata frameworks, and data integration systems supporting institutional analytics.
- (iii). Establish and monitor automated data ingestion, cleaning, transformation, and validation pipelines for diverse scientific, clinical, and laboratory data streams.
- (iv). Design interactive data visualizations, surveillance dashboards, and decision-support tools for real-time tracking of disease outbreaks and field research metrics.
- (v). Provide advanced, data-driven monitoring and evaluation analytics to assess research project performance, therapeutic trial outputs, and scientific programmatic interventions.
- (vi). Coordinate the secure dissemination and deployment of analytical outputs through institutional data portals, technical reports, and integrated scientific publishing frameworks.
- (vii). Implement logical data safeguarding protocols, secure encryption matrices, and automated data curation profiles across all analytical nodes.

4. Staffing

The Data Science and Analytics Division is staffed as follows in the table below

Designation	Grade KIP	Approved Establishment	In post	Proposed Establishment	Justification
Deputy Director, Data Science & Analytics	3		0	1	To head and lead the division
Chief Research Scientist	2		0	1	To lead resource mobilization and high Level modelling.

Senior Principal Research Scientist I	3		0	1	Spearhead advanced modelling and institutional technical mentorship
Sub-total			0	3	Core divisional management team
Biostatistics and Computational Analytics Section					
Assistant Director	4	1	0	1	Provide technical leadership, oversee section workflows, and coordinate institutional data portfolios.
Senior Principal Research Scientist II	4	2	0	2	Supervise complex study designs, multi-entity biostatistical reporting, and advanced computational pipelines.
Principal Biostatistician / Bioinformatician / GIS Specialist / Research Scientist	5		0		Retained for career progression: Lead multi-omics integration and architect advanced geospatial and epidemiological models.
Senior Biostatistician / Bioinformatician / GIS Analyst / Research Scientist	6		0		Retained for career progression: Design bioinformatics pipelines, manage GIS platforms, and map disease distribution.
Biostatistician / Bioinformatician / GIS Officer / Research Scientist II/I	8/7	2	2	2	Execute data cleaning protocols, deploy GIS scripts, and provide daily clinical/epidemiological study support.
Sub Total		5	2	5	Manage pre-clinical studies and epidemiological workloads
Data Analytics and Decision Support Section					
Assistant Director	4		0	1	To head the section

Senior Principal Research Scientist II	4		0	2	To implement logical data safeguarding protocols and architect secure automated AI enabled cleaning workflows
Principal Data Scientist/Research Scientist	5		0		Retained for career progression
Senior Data Scientist/Research Scientist	6		0		Retained for career progression
Data Scientist/ Research Scientist II/I	8/7		0	2	To administer automated ingestion pipelines and index reference artifacts
Sub Total			0	5	To protect data integrity, automate verification paths, and Secure institutional research memory
Total		5	2	10	To fully operationalize the independent division, correcting critical technical deficits across All three pillars.